

Measuring the Distance to the Dark Matter-less Galaxy NGC 1052-DF2

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Abstract

Most galaxies are believed to be contained in dark matter halos, however, a few have recently been found as potential exceptions. One ultra diffuse galaxy in particular, NGC 1052-DF2 (DF2), has had controversy over whether or not the visible matter in the galaxy can account for all of the velocity dispersion in the galaxy. Some measurements find that there is a deficiency of dark matter at a larger distance $\gtrsim 16$ Mpc, while others find that with a smaller distance ~ 13 Mpc there is still the need for dark matter to explain the physics. Several different types of distance measurements have been made in attempts to settle whether or not DF2 lacks dark matter. We measure the distance using 16 globular clusters (GCs) detectable within the galaxy and thus fit GC luminosity function (GCLF). From this fit, we use known empirical relations to measure distance from the absolute turnover magnitude using the distance modulus. We use the Hubble Space Telescope (HST)/Advanced Camera for Surveys (ACS) observations of DF2 from 2020 to estimate a distance of $d = 16.3 \pm 1.0$ (stat) ± 1.6 (sys) Mpc. By providing an independent measurement of the distance to this ultra diffuse galaxy, we find tentative evidence to support other works that show that DF2 does not require dark matter for rotational support.

1 Introduction

Galaxies have generally been found to contain dark matter halos, with significantly more dark matter than baryonic matter. Within the last decade, at least two galaxies have been identified that lack enough dark matter to fit other observations from galaxy surveys (Van Dokkum et al., 2018b). In particular, the diffuse galaxy DF2 has been debated over whether or not its dark matter content is consistent with other galaxies due to its distance. While originally placed at around 20 Mpc, similar to NGC 1052 (Van Dokkum et al., 2018b), Trujillo et al. (2019) makes several independent measurements (including the method used in this work) and finds from averaged measurements that DF2 is closer than previous estimates and thus the inferred dark matter absence is accounted for instead by a distance of 13 Mpc. Subsequent works (e.g. Danieli et al. (2020); Shen et al. (2021, 2023); Beasley et al. (2025)) have made various further distance estimates using numerous methods, including the tip of the red giant branch, H₂O mega-MASER distance ladders, globular cluster velocity dispersion relations, and surface brightness fluctuations. Therefore, in measuring this distance independently and replicating one of the several methods used by various groups, we provide an important contribution to work towards settling whether or not DF2 is truly lacking in dark matter. That is, does DF2 require the presence of dark matter to explain its physical properties?

The contending measurement of ~ 13 Mpc is from a combined distance estimate with five methods (Trujillo et al., 2019) while further distance measurements of between $\sim 16 - 20$ Mpc are consistent with a dark matter deficiency and inconsistent with the former distance (at 1σ uncertainty) (Fahrion et al., 2025; Van Dokkum et al., 2018a,b). The method used here to measure distance is motivated by work that finds a “standard candle” in the histogram of magnitude measurements of all GCs in a galaxy Hanes (1977); Harris (1991); Rejkuba (2012). Thus, using a known absolute turnover magnitude fit for a gaussian distribution of GC magnitudes, we can estimate the distance. However, recent work (Shen et al., 2021) finds a more complex result of a combined two-gaussian fit for the GCLF indicating two distinct subpopulations of GCs. In that analysis, though, they assume a known distance modulus from the literature and do not use the apparent magnitudes to calculate one. Thus, there is strong motivation to use the fit to calculate the distance independently with data which has not been used with this method to compare to other distance measurements in hopes of settling the dark matter issue of DF2.

2 Data

We use the HST observations of DF2 from Proposal 15851 (PI van Dokkum).¹ The observations made between December 2020 and March 2021 are fully reduced, drizzled, and mosaicked images with filenames `hst_skycell-p1064x11y18- _acs_wfc_XXXX_all_drc.fits` where XXXX is the respective broad I (F814W) and V (F606W) filters. These were downloaded from the MAST astronomical data archive via the Astroquery Python package (Ginsburg et al., 2019). The images were taken with the HST ACS instrument over 38 orbits. These specific data products are used in this work because they are more up-

¹https://archive.stsci.edu/proposal_search.php?id=15851&mission=hst

dated observations than those from [Van Dokkum et al. \(2018b\)](#) and are of DF2, which is a galaxy that is believed to contain little dark matter. Since many measurements have been made that are inconsistent with one another (see Fig. 5 in Sec. 4), performing a distance estimate with a different method with some of the best data is important. Before beginning the analysis, we take a 3' cutout centered on DF2. The data at this stage is shown in Fig. 1 with the GCs marked.

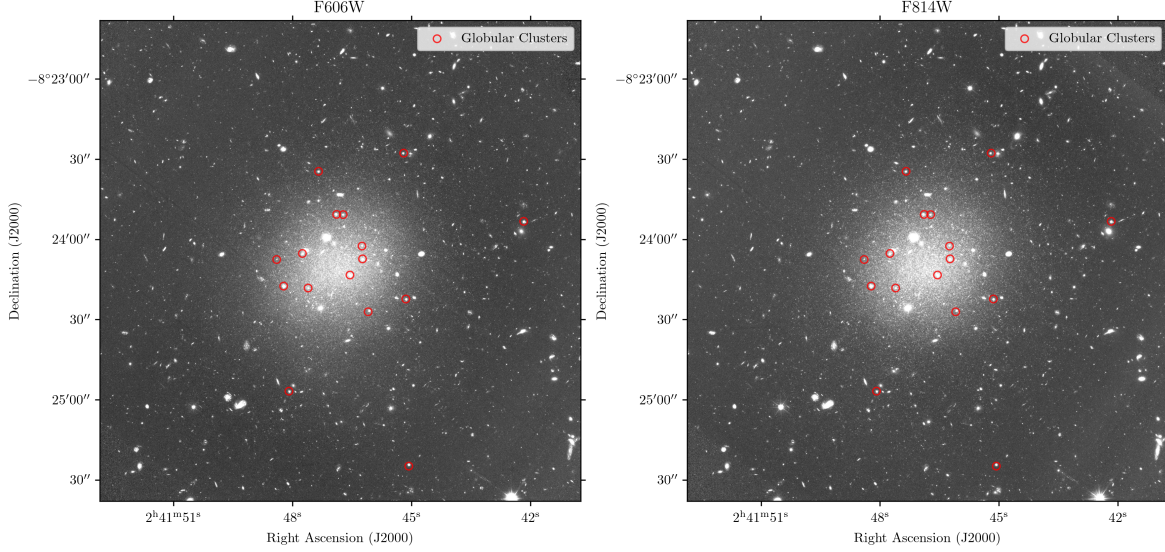


Figure 1: NGC 1052-DF2 in F606W (left) and F814W (right) filters from the HST ACS. Globular clusters listed in Tab. 4 are circled in red in each image. Each image is normalized with an IRAF zscale for visualization.

3 Methods and Analysis

For both images, F606W and F814W, the procedure is the same as follows. We use the right ascension and declination of GCs from the table in [Fahrion et al. \(2025\)](#) as initial position estimates then use the `quadratic_centroid` function from the `photutils` package to find the true center of each GC by fitting a 2D quadratic polynomial ([Astropy Collaboration et al., 2013, 2018, 2022](#)). In order to perform accurate photometric measurements, we must subtract excess flux counts added by background sky and the galaxy. We estimate the background, including the sky and that caused by the diffuse galaxy light itself, and subtract it with `photutils Background2D`, a grid interpolated estimate using a `MedianBackground` estimator. Fig. 2 shows the background and galaxy subtracted images for the V and I filters used for photometry.

To perform the necessary photometric calculations to obtain apparent magnitudes, we first measure the summed counts at a fixed aperture radius of 6 pixels for each GC with the `photutils aperture_photometry` method. Then, we correct for the known encircled energy fractions for the respective filters at this radius (roughly 0.2''), as given by [Bohlin \(2016\)](#). Finally, the corrections to flux for dust extinction A_λ are made using the values from NASA/IPAC infrared science archive for the appropriate V and I filters. We use Vega zeropoints to convert to magnitudes from fluxes.

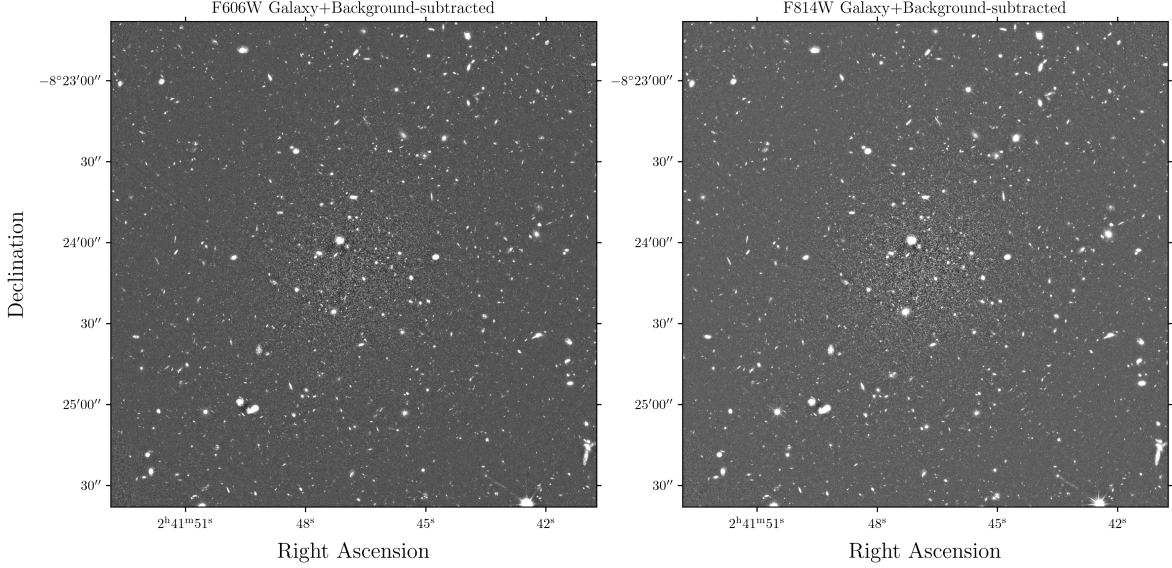


Figure 2: Same as Fig. 1 but background and galaxy subtracted.

To then fit a GC luminosity function with a gaussian, we perform an unbinned maximum likelihood estimation function using the `scipy` Nelder-Mead `minimize` method. This allows us to not lose information by binning magnitudes into a histogram since we only have 16 GCs, and instead fits many gaussian functions with different parameters and finds the parameters with the maximum likelihood of fitting the data:

$$\mathcal{L}(\mu, \sigma) = \prod_{i=1}^{16} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(m_i - \mu)^2}{2\sigma^2}\right). \quad (1)$$

To then account for statistical error in this fit with such a small sample size, we bootstrap the gaussian fit 1000 times introducing small, random fluctuations in the magnitudes corresponding to the uncertainties in the magnitudes calculated previously. We then take the median and 16th and 84th percentiles to obtain the final apparent magnitude measurement and 1σ uncertainties.

To also attempt to fit two gaussian functions, as is potentially the case with this GCLF indicated in [Shen et al. \(2021\)](#), is even more difficult with this small dataset as there are five degrees of freedom: $\Lambda = [\mu_1, \mu_2, \sigma_1, \sigma_2, \mathcal{M}]$ (where the last is the mixing fraction). The procedure is the same unbinned maximum likelihood estimation, just now with two gaussian distributions weighted by $\mathcal{M}$:

$$\mathcal{L}(\Lambda) = \prod_{i=1}^{16} \mathcal{M}p(\mu_1, \sigma_1) + (1 - \mathcal{M})p(\mu_2, \sigma_2). \quad (2)$$

We analyze the corrected Akaike information criterion to determine if one or two gaussian components better describe the data given the limited degrees of freedom.

Using the turnover magnitudes and the absolute magnitude relation for GCLFs [Rejkuba \(2012\)](#), we estimate the distance:

$$d = 10^{\frac{m-M+5}{5}}, \quad (3)$$

Name	RA	Dec	m_{F606W} (Vega mag)	m_{F814W} (Vega mag)
GC-39	02:41:45.07	-08:25:24.3	23.1258 ± 0.0009	22.0659 ± 0.0005
GC-59	02:41:48.08	-08:24:56.9	23.410 ± 0.001	22.4515 ± 0.0007
GC-71	02:41:45.13	-08:24:22.4	23.378 ± 0.001	22.3041 ± 0.0006
GC-73	02:41:48.22	-08:24:17.6	22.2217 ± 0.0004	21.2114 ± 0.0002
GC-77	02:41:46.54	-08:24:13.5	22.9937 ± 0.0008	21.7648 ± 0.0004
GC-85	02:41:47.75	-08:24:05.4	23.0256 ± 0.0008	21.9951 ± 0.0005
GC-91	02:41:42.17	-08:23:53.4	23.366 ± 0.001	22.1861 ± 0.0005
GC-92	02:41:46.90	-08:23:50.6	22.7286 ± 0.0006	21.7861 ± 0.0004
GC-93	02:41:46.73	-08:23:50.8	23.436 ± 0.001	22.5072 ± 0.0007
GC-98	02:41:47.35	-08:23:34.6	23.413 ± 0.001	22.4418 ± 0.0007
GC-101	02:41:45.21	-08:23:27.8	23.518 ± 0.001	22.5325 ± 0.0008
GCNEW2	02:41:47.60	-08:24:18.3	24.484 ± 0.003	23.371 ± 0.002
GCNEW3	02:41:46.23	-08:24:07.4	23.320 ± 0.001	22.2828 ± 0.0006
GCNEW4	02:41:48.40	-08:24:07.4	24.472 ± 0.003	23.724 ± 0.002
GCNEW5	02:41:46.25	-08:24:02.3	24.803 ± 0.004	23.498 ± 0.002
GCNEW9	02:41:46.06	-08:24:26.7	24.772 ± 0.004	23.246 ± 0.001

Table 1: GCs used in this analysis along with accompanying apparent magnitude measurements in the V and I filter bands in this work. We use new GCs identified in [Fahrion et al. \(2025\)](#) and previous ones found in ([Van Dokkum et al., 2018b](#)).

where m is the apparent magnitude calculated, M is the known absolute turnover magnitude, and d is the distance in pc.

4 Results

In Tab. 4, we list the apparent magnitudes in the Vega system calculated using the methods described in Sec. 3 for all 16 GCs analyzed. Using these apparent magnitudes, we fit gaussian distributions for the GCs in each filter and following the procedure described above, bootstrap these fits 1000 times with perturbations normally distributed around 0 with σ corresponding to the uncertainties in Tab. 4. We use the absolute turnover magnitudes derived by [Rejkuba \(2012\)](#) from the empirical fitting of normal distributions to GC populations as $M_V^{TO} = -7.66 \pm 0.09$ mag and the corresponding $M_I^{TO} = -8.46 \pm 0.09$ mag. The result are shown in Fig. 3, where we find a median distance in the V band to be $d_{F606W} = 17.3^{+1.4}_{-1.3}$ (stat) ± 1.6 (sys) Mpc, in the I band to be $d_I = 15.3^{+1.2}_{-1.1}$ (stat) ± 1.6 (sys) Mpc and the mean of the two combining all bootstraps from both bands to be $d = 16.3 \pm 1.0$ (stat) ± 1.6 (sys) Mpc. The systematic uncertainty comes from the propagation of uncertainty from the magnitude conversions (0.09) into a distance, while the statistical uncertainty comes from the 16th and 84th percentiles of the bootstrapping. The measurements in both filters are consistent at the 1σ level even if including only statistical or systematic uncertainty. The measurements are also broad in their total uncertainty range, a quadrature sum of ± 1.9 Mpc, which stems from the small sample size of GCs and systematic uncertainty in the GCLF absolute magnitude turnover. The mean distance measurement is consistent at $< 1\sigma$ with all but the two most extreme measurements from the literature, as shown in Fig. 5 ([Trujillo](#)

et al., 2019; Shen et al., 2023).

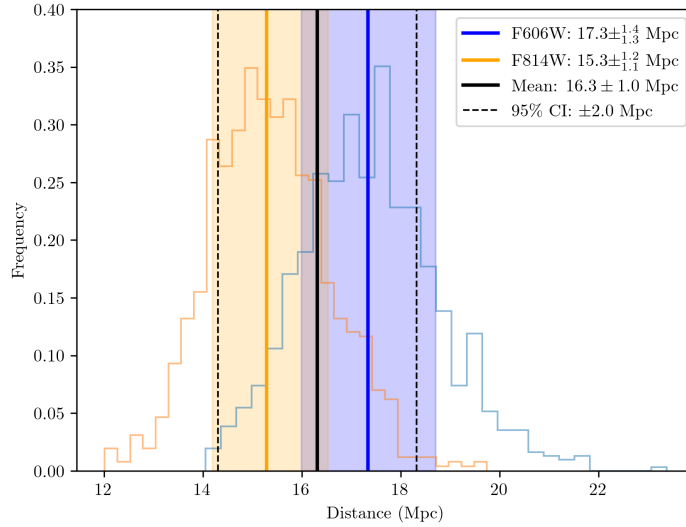


Figure 3: Normalized histogram of bootstrapped calculations of the distance to DF2 for the F606W filter (blue) and the F814W filter (yellow). Shaded regions correspond to uneven 1σ bounds for the respective filters. The solid black line marks the mean measurement of distance, and the dashed black lines mark the 95% confidence interval for combined bootstrapped calculations. The solid blue and yellow lines correspond to the median distances of the filters and to the 1σ interval of the mean distance. Uncertainties here are only statistical, and bootstrapping is performed for magnitudes and then those are converted to distances.

To check whether an alternative, more complex GCLF better models the data such that there may be two distinct GC subpopulations, we also follow the procedure described in the above section to calculate a mixture of two gaussian distributions. These, alongside the single gaussian, are shown in Fig. 4 placed on top of histograms of the magnitudes for both the V (left) and I (right) bands. The sample size poses a significant challenge in fitting the 5 degrees of freedom in a gaussian mixture model (see Eq. 2), and to check whether the double or single gaussian model fits the data best, we calculate the Corrected Akaike Information Criterion Hurvich & Tsai (1993) (cAIC) for small sample sizes. While the absolute magnitudes are not informative, the smaller relative magnitude marks which model is most appropriate for the data, considering sample size as well. We find the cAIC for the double gaussian (44.99 and 46.85 for V and I respectively) is larger than the single gaussian (40.15 and 38.01) and, therefore, do not report any distance calculations with peaks of the potential subpopulations from the mixture model. In addition, for that reason there is not significant statistical support for using a more complex GCLF to fit this data.

5 Conclusions

In this work, we make an independent distance measurement using the GCLF turnover of DF2 of $d = 16.3 \pm 1.0$ (stat) ± 1.6 (sys) Mpc. This measurement is made using HST/ACS observations in the V and I bands and 16 previously identified GCs. The

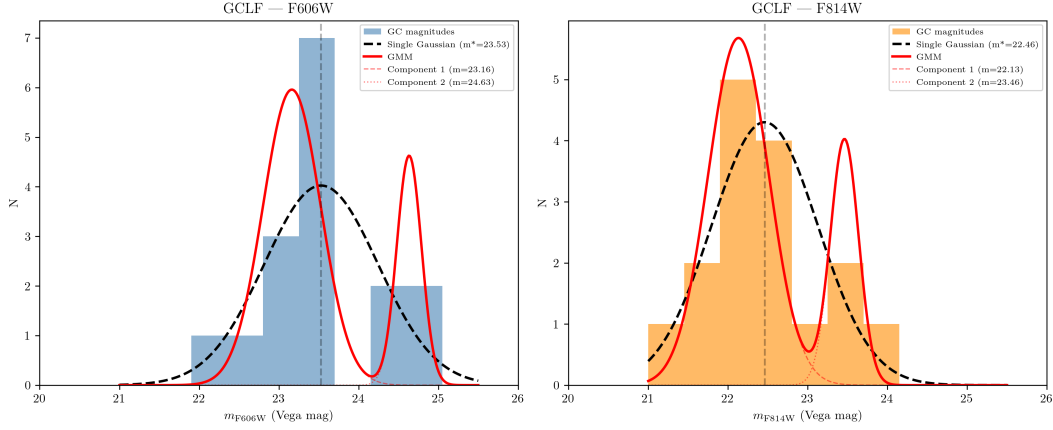


Figure 4: Comparison of the single and double gaussian models for the histogram of apparent Vega magnitudes for both V (left) and I (right) filters. The median magnitudes of the GCLFs are marked in the vertical dashed lines.

16.3 Mpc distance estimate is exceptionally consistent with the recent estimate made using globular cluster velocity dispersions of 16.2 Mpc (Beasley et al., 2025). It is also consistent with most other estimates made in the literature at $< 1\sigma$ apart from the 13.0 Mpc (Trujillo et al., 2019) and 21.7 Mpc measurements. However, due to the combined systematic and statistical uncertainties, the distance measurement in this work should not be taken as a definitive demonstration that there is no dark matter in DF2, as other factors can play a role (Beasley et al., 2025). We do present evidence, however, that the distance estimate is more consistent with those that are able to account for no dark matter in DF2 as opposed to those that confidently allow for dark matter in the ultra diffuse galaxy. Lastly, we are unable to present statistically significant evidence that the GCLF is composed of more than one underlying GC subpopulation. With new data from JWST, more precise and accurate measurements of the distance to DF2 may be made, enabling a settling of this dark matter controversy.

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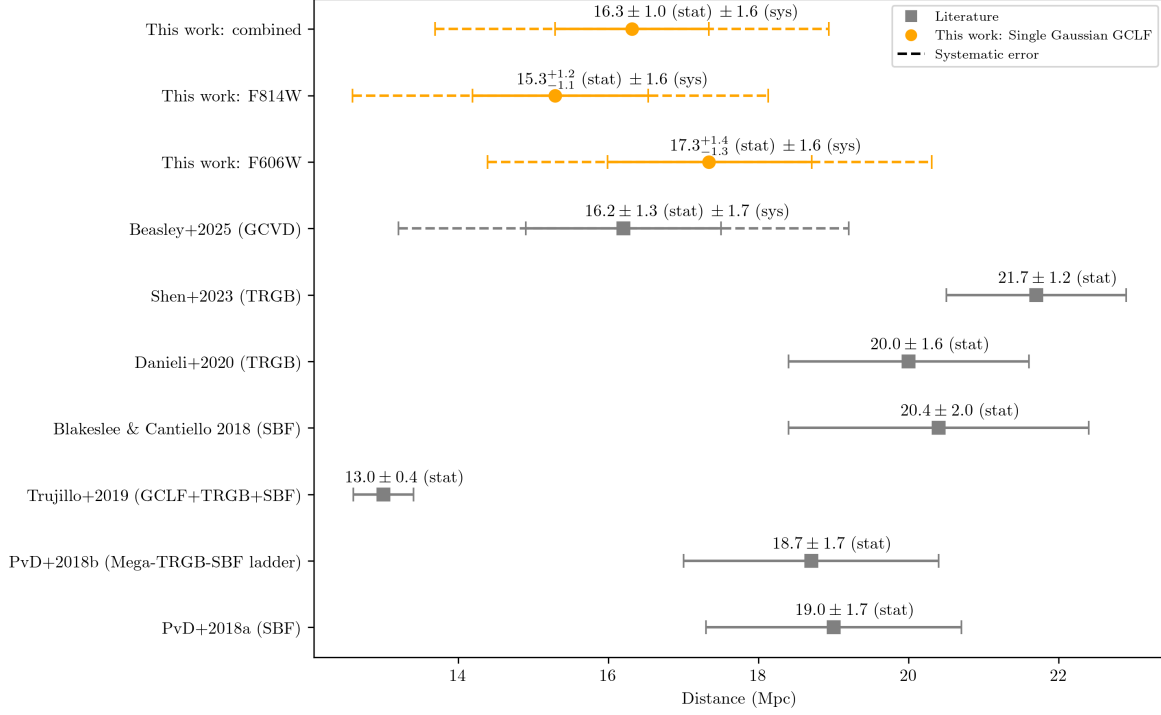


Figure 5: Comparison of distance estimates to NGC 1052-DF2 from this work and the literature. Orange circles show the GCLF-based distances derived in this work using unbinned maximum likelihood fitting of a single Gaussian to the F606W, F814W, and combined magnitude distributions of 16 globular clusters. Solid error bars represent statistical uncertainties from bootstrap resampling; dashed error bars include the additional systematic uncertainty from the absolute calibration of the GCLF turnover (Rejkuba, 2012). Gray squares show literature distance measurements using a variety of methods as labeled (Van Dokkum et al., 2018b,a; Blakeslee & Cantiello, 2018; Trujillo et al., 2019; Danieli et al., 2020; Shen et al., 2021, 2023; Beasley et al., 2025).

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